CATIA USER SYMPOSIUM AMERICAS 2026

MAY 18-21 | RENO | NEVADA | USA

CUSA 2026 · BAE SYSTEMS & ENOLA TECHNOLOGIES

A DIGITAL ENGINEERING BRIEFING

HIDDEN IN PLAIN SIGHT

Data Aggregation Risks Across the Digital Engineering Landscape

BAE SYSTEMS

ENOLA

OUTFOX THE COMPETITION

Patrick Rose · BAE Systems

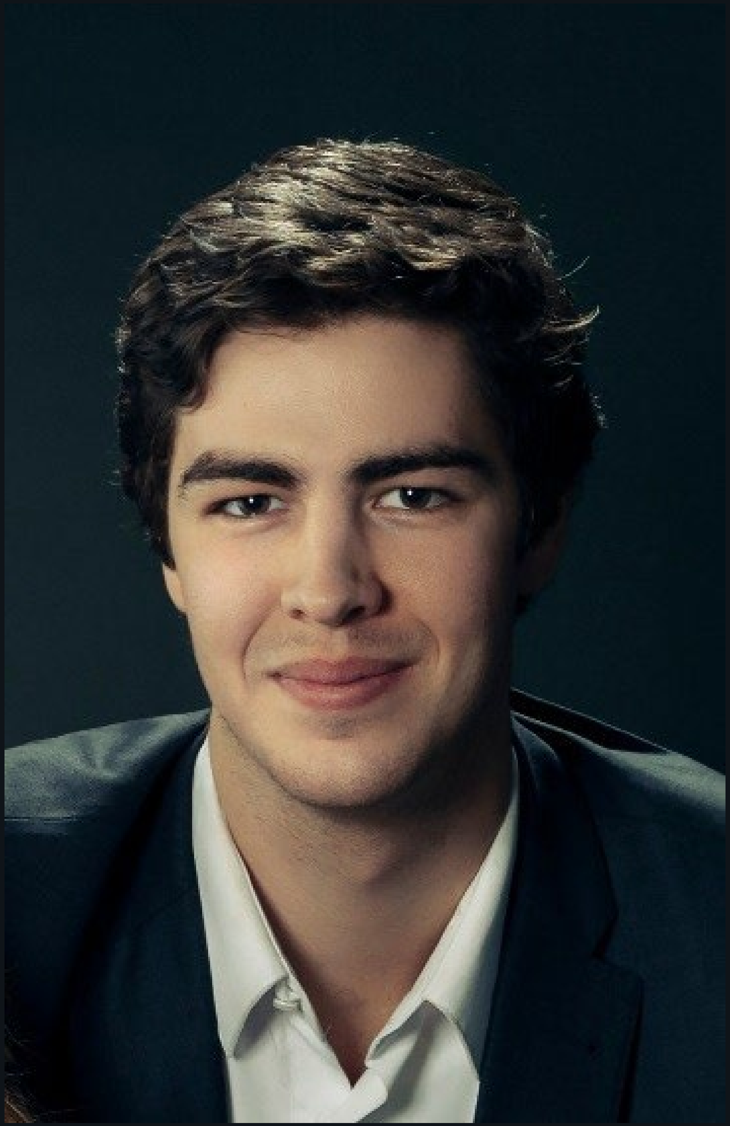
David Fields · Enola Technologies

PATRICK ROSE

Principal Systems Engineer · BAE SYSTEMS

5+ years experience defining systems-of-systems architecture for DoD programs including transformation of 30+ years of legacy system definition.
Builds architectures and external tool integrations to drive automated M&S efforts.

MBSE | CATIA MAGIC | SYSML | M&S | Transformation



KEY EXPERIENCE

- MDA-AB System Architecture Lead · BAE Digital Engineering Lead
- [Notable role / responsibility]
- Specialties: Systems architecture, automation, legacy to digital data transformation
- BS Mechanical Engineering – Johns Hopkins University

PRESENTER / 02

DAVID FIELDS

Co-Founder / Chief Technology Officer · ENOLA TECHNOLOGIES

20+ years of MBSE consulting and training across DoD digital engineering programs. Builds the tooling and training behind Enola Technologies from intro SysML to advanced MagicDraw API and simulation work.

TWC | AUTOMATION/API | SYSML v1/v2 | DSL | AI/LLM/MBSE

KEY EXPERIENCE

- NAVAIR · MQ-25 MBSE Lead · SET Functional Lead · Naval IME Lead
- Army PEO Aviation · FLRRA Digital Engineering Lead
- Specialties: automation, profiling & DSL, enterprise MBSE deployment
- BS Computer Engineering - University of Arkansas

TODAY'S ROADMAP

01
THE PROBLEM IN PLAIN TERMS
Current policy and procedures don't align with digital engineering artifacts.

02
WHY THIS ISN'T ONLY A SECURITY ISSUE
IP, ITAR/EAR, HIPAA have the same pattern, different regulator

03
WHERE THE GUIDANCE FALLS SHORT
SCGs, NISPOM, and vendor docs were built for static documents

04
LIVE DEMO · CATIA MAGIC / MAGICDRAW
Three classification cases that current tooling cannot resolve

05
WHERE THE CONVERSATION NEEDS TO GO
What we want from vendors, government, and the community

06
QUESTIONS & DISCUSSION
We don't have answers... we want yours!

THE BOTTOM LINE

We don't have a fix. We have a problem, and a conversation that needs to start.

INDUSTRY-WIDE

No clean answer today across vendors, government, or programs.

NOT JUST CLASSIFICATION MARKING

IP, ITAR / EAR, HIPAA have the same combinatorial leak pattern.

GUIDANCE GAP

Marking standards were written for documents, not digital tools or ecosystems.

SHARED LIFT

Vendors, federal government, and industry need to pull together.

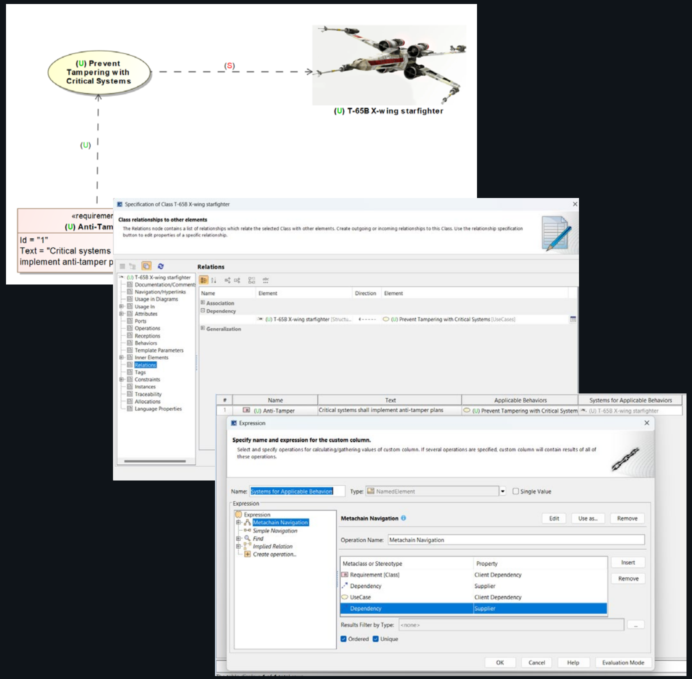
WHERE THE DATA ACTUALLY LIVES

- Engineering data lives in models, repositories, simulations, and dashboards not just documents

- A single element can carry different classifications across views, columns, and aggregations
  - e.g. a Secret requirement with a CUI title: title view ≠ body view

- Current SCGs and marking guides were written for page-and-paragraph artifacts

- "Mark everything to the highest level" over-classifies and blocks collaboration



IT'S BIGGER THAN CLASSIFICATION

SECURITY

Security Classification

CUI · FOUO · Secret · TS

- Aggregation rules turn unclassified parts into classified wholes
- Views, columns, and joined tables lose the marking context
- Cross-domain solutions inherit the ambiguity downstream

IP / EXPORT

IP & Export-Controlled

Trade secrets · ITAR · EAR

- Third-party IP sits next to in-house IP in the same model
- ITAR / EAR controlled technical data has the same combinatorial risk
- Jurisdictional consequences amplify a leak

PRIVACY

Privacy & Regulated

HIPAA · GDPR · PII · PHI

- De-identified fields can re-identify in joined views
- Identical digital aggregation pattern but a different regulator
- Audit and breach reporting frameworks all assume documents

BUILT FOR PAPER. USED FOR CODE.

DODM 5200.01 · NISPOM · PROGRAM SCGS

All written around portion-marked documents.

Tool vendor docs and training mirror the same assumption.

- No widely adopted standard for marking model elements, source code, relational data, tool UI/UX, dashboards
- Engineers improvise while auditors have no benchmark and programs absorb the risk
- Tooling defaults to "highest classification wins," which over-classifies by construction
- Training pipelines still drill page paragraphs, pages, and documents not artifacts
- We're not training people for the artifacts they produce

THREE SAMPLES IN CATIA MAGIC

- Walking through three sample classification rules and the challenges each one surfaces
  - Demonstrated in CATIA Magic / MagicDraw using the Data Markings Plugin
- All classification markings shown are notional and for demonstration only
- Demo model and slides will be available for download after the session
- Goal: spark the discussion - **we are not prescribing a solution**

DEMO MODEL · DOWNLOADABLE

Working .mdzip + slides at the URL on the wrap-up slide.

CASE OVERVIEW

01

Aggregation

When unclassified parts add up to a classified whole

- Combinations of unclassified parts cross a classification threshold
- Custom columns silently traverse classified relationships
- Dynamic views and tool dialogs do not surface the resulting marking

02

Instantiated Values

A property is fine until it has a value

- A property is unclassified until it is populated
- No clean way to mark "becomes classified once a value is assigned"
- Marking the value itself does not appear in diagrams or dialogs

03

View Layers

Smart packages and legends rewrite the marking

- Smart packages, legends, and filters change a view at render time
- No notification that the resulting view has a different classification
- Same data, different view → different effective marking

LIVE DEMO

CATIA MAGIC / MAGICDRAW · CLASSIFICATION CHALLENGES IN PRACTICE

Three cases · No solution offered · Discussion to follow

WHAT WE JUST SAW

01 SPECIFICATION WINDOW

The Specification Window does not reflect the overall classification based on the currently visible data.

02 QUERIES

Custom scripts and queries may traverse classified data without inheriting the marking.

03 PROPERTY VALUES

Property values change a property's effective classification the moment they are populated.

04 DYNAMIC VIEWS

Smart packages and legends silently re-classify a view at render time.

05 AMBIGUITY

The same model produces different classification answers depending on how you ask the question.

EXTENDING THE DISCUSSION

SYSML V2 / KERML
Textual notation adds additional complexity to the problem.

LARGER DE ECOSYSTEMS
PLM, ALM, requirements, simulation, analytics all share the problem.

CROSS-DOMAIN SOLUTIONS
CDS pipelines that must make marking decisions automatically.

WHAT WE NEED FROM EACH OF YOU

VENDORS

Tool Vendors

Make data marking a first-class concept

- Native, view-aware classification semantics
- First-class marking concept in the metamodel and the UI
- Surface the marking impact of every view, column, and query

GOV / DOD

Federal Govt / DoD

Update guidance for digital artifacts

- Update SCGs, DoDM 5200.01, and program guidance to address digital artifacts directly
- Fund reference implementations and pilot programs
- Engage with vendors on standards, not just policy

INDUSTRY

Industry & Community

Build and share what works

- Reference patterns for marking digital artifacts
- Refresh training to cover models, code, and dashboards not only documents
- Bring program lessons learned back to the community

CONCLUSION

We don't have a complete answer ...and that is exactly the point.

EVERY DOMAIN

The problem touches every regulated information domain not just security classification.

GUIDANCE GAP

Marking guidance was built for documents. Digital engineering needs guidance built for digital artifacts.

SHARED LIFT

Vendors, government, and program teams have to engage on this together.

TAKE & SHARE

Take what fits your program today, and bring your lessons back to the community.

QUESTIONS ?

STAY IN TOUCH

Patrick Rose · BAE Systems · *patrick.rose2@baesystems.us*

David Fields · Enola Technologies · *david.fields@enolatech.com*

GET THE BRIEF & MODEL

SCAN THE CODE OR FOLLOW THE LINK

github.com/EnolaTechnologies/cusa26

WHAT'S INSIDE

- This brief — slides (.pdf)
- Demo model — working .mdzip for MagicDraw / CATIA Magic 2024x
- Other Enola sessions from CUSA 2026

STAY IN TOUCH

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